

Retrospective Confirmation for Identity-Consistent Streaming 3D Perception

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Abstract

001 *Streaming 3D perception systems emit objects whose identities and class labels flicker: the same physical object is*
002 *re-detected as a new instance, or changes class, from one*
003 *frame to the next. We present a training-free retrospective*
004 *confirmation module that runs on frozen detector output and*
005 *changes no box coordinate, score, or label. A track is emitted*
006 *only after N observations, the standard confirmation*
007 *test; at confirmation the module retrospectively releases the*
008 *withheld prefix, a step online trackers omit, recovering the*
009 *trajectory whole. Because such a rule also reduces how*
010 *much a system emits, and most metrics respond to volume*
011 *alone, we validate it against matched controls: a score*
012 *threshold at the identical detection count, and a top- K selection*
013 *at the identical identity count. On nuScenes with a*
014 *frozen detector, association accuracy improves against every*
015 *control in both association frames (AssA +0.049 in the*
016 *sensor frame), with bootstrap intervals excluding zero. On*
017 *ScanNet200 — an indoor open-vocabulary pipeline sharing*
018 *no detector, modality, or codebase — the same module*
019 *reduces class-label switching by 26.9% against an identity-*
020 *matched control, with segmentation accuracy unchanged.*
021 *The gain is in consistency specifically, and it is paid for in*
022 *detection recall and in $N-1$ frames of latency.*
023

024 1. Introduction

025 Streaming 3D perception systems — detectors and semantic
026 mappers that run on a live sensor feed [10–12, 16] —
027 are consumed downstream as *persistent objects*: a map, a
028 scene graph, a planner’s object memory. What those consumers
029 need is that one physical object stay one object
030 with one label. What streaming systems actually produce
031 flickers, because the prevailing emission rule is a per-frame
032 confidence threshold: every detection above a score cut-off
033 enters the stream immediately, independently of what
034 was emitted a frame earlier. A re-detected object arrives
035 as a new identity, and an unstable open-vocabulary label
036 changes class along the way. Per-frame detection metrics
037 do not see this, and so do not penalise it. Temporal reme-

dies exist — confirmation counts, and confidence accumulated
along a track [1, 19, 20, 22] — but each also reduces
how much a system emits, and detection-oriented metrics
are strongly coupled to output volume, so a comparison that
does not hold that volume fixed reports what a rule *selects*
and how much it *removes* together. The property is thus
both unmeasured by the dominant metrics and unattributed
by the dominant comparison.

We study a *retrospective confirmation* module in that setting,
requiring no retraining and no change to the perception
model. Detections are linked into tracks by a class-agnostic
associator; a track becomes *confirmed* only after N observations,
and at that moment the module emits not only the current
detection but the track’s withheld prefix, so a confirmed
object’s trajectory is complete from its first observation.
Nothing about a box’s coordinates, score, or label is modified,
and the module runs at a single operating point ($N=3$)
with no per-dataset or per-metric tuning.

The attribution problem applies to our module as much
as to any other, so we evaluate under *matched controls*
that hold the output budget fixed while the selection rule
varies. A *detection-budget-matched* control thresholds the
same frozen detector output to emit exactly as many detections
as the module; an *identity-budget-matched* control keeps,
per sequence, the highest-scoring K tracks, where K is the
number of identities the module emits. To compare persistence
against the other established temporal rule rather than against
score alone, a third control accumulates confidence along a
track in the manner of CBMOT [1] and is held to both
budgets. All controls share the detector, the association step,
and the evaluation code with the module, differing only in
the rule that selects what is emitted.

Under these controls, on the nuScenes validation set (150
scenes, a frozen CenterPoint [36] detector, both sensor-frame
and ego-motion-compensated world-frame association, confirmed
tracks emitted retrospectively), the module improves association
accuracy (HOTA’s AssA component [21]) over both matched
controls in both association frames, with bootstrap confidence
intervals excluding zero, while class-aware tracking accuracy
AMOTA [6] — a point estimate, since it does not admit the
same per-scene resam-

pling — rises consistently in sign and magnitude. At the same output budget it *reduces* per-frame detection quality (Sec. 5.1). The gain is bought rather than added.

Against confidence accumulation this becomes the paper’s central observation: at a matched budget the two temporal rules do not order one another but sit at different points of the same trade-off. Persistence yields higher association accuracy, accumulation retains the advantage on the detection- and identity-oriented measures, and each of these differences has an interval excluding zero. The separation is the same size whether emitted *detections* or emitted *identities* are held fixed, which rules out the simplest explanation — that persistence gains association accuracy merely by carrying fewer identities to confuse.

Because the module touches only emission timing, it transfers to any streaming pipeline that re-observes objects. We apply the identical module, unchanged and at the same N , to an indoor open-vocabulary instance-segmentation pipeline that shares no detector, modality, or codebase with the outdoor one, and evaluate it under the same control design. On ScanNet200 (312 validation scenes, 3D instance-mask proposals labeled per frame by a 2D open-vocabulary detector), the module reduces the number of frames in which an instance’s class label changes by 26.9% relative to an identity-budget-matched control of equal size, while leaving instance-segmentation AP essentially unchanged. The temporal system has fewer label switches in 309 of 312 scenes and the paired confidence interval excludes zero by a wide margin, whereas selecting the same number of instances at random reproduces none of the reduction — so the gain is specific to temporal selection rather than to emitting fewer identities.

Contributions.

- **A training-free retrospective confirmation module.** An operator combining the standard confirmation test with retrospective emission of the confirmed prefix, applied to frozen detector output at a single operating point, with no retraining, no architectural change, and no modification of box coordinates, scores, or labels.
- **The retrospective emission policy, isolated.** We show the retrospective step is what completes the module: emitting causally instead narrows *where* the gain holds — it survives in the sensor frame but reverses in the world frame — and retrospection’s price is $N-1$ frames of latency (1.0 s at 2 Hz for $N=3$).
- **A matched-budget comparison of two temporal selection rules.** Confirmation and confidence accumulation are compared at an identical emission budget and again at an identical identity budget, with paired per-sequence statistics. Neither dominates: persistence buys association accuracy, accumulation keeps the detection- and identity-oriented measures, and the separation is unchanged under either matching.

- **Controls that separate a rule from its suppression.** Both budgets are also matched against a non-temporal score threshold, and indoors against random selection, so what the comparisons credit is the selection rule rather than the smaller volume it emits. The design is not specific to confirmation and applies to any component that alters output volume.
- **Cross-domain validation under that design.** Outdoors, the module improves association accuracy and class-aware tracking accuracy under both controls in both association frames, at a detection cost we quantify in Sec. 5.1 rather than leave implicit. Indoors, the identical module reduces class-label switching on an unrelated open-vocabulary instance-segmentation pipeline, with segmentation quality unchanged.

2. Related Work

Streaming perception. Streaming perception was formalised by [16], which scores predictions against the world at the moment they are *emitted*, folding latency and accuracy into streaming AP (sAP); later work optimises detectors for this regime [35], and the 3D analogue processes LiDAR sectors as they arrive rather than waiting for a full sweep [11]. This line establishes our premise but corrects *when* a prediction is scored: sAP remains a per-frame average precision, so two systems emitting identical boxes at identical latency but differing entirely in identity continuity and label stability receive the same sAP. We hold latency and box geometry fixed and ask what that family cannot see.

Track lifecycle and confirmation. Deciding when a candidate track is real enough to report is among the oldest problems in multi-target tracking. The classical radar literature settles it with *track confirmation* — under M -out-of- N history logic a track is confirmed once it has been assigned a detection in M of the last N updates, under score logic once its likelihood ratio crosses a sequential threshold — and designs confirmation, deletion, and track management together [4]. The same literature established that reporting may lag estimation: *retrodict* revises earlier estimates once later measurements arrive, in exchange for a bounded delay [15]. Tracking-by-detection inherited the count-based half: SORT [3] and DeepSORT [33] withhold a tentative track until matched in N consecutive frames, and AB3DMOT [32] carries the same minimum-hits rule onto 3D boxes, while CenterPoint [36] keeps a max-age deletion rule but reports tracks from their first observation. None of this is ours: the confirmation test we use is the textbook one at its conventional operating point. Our retrospective emission likewise buys completeness with delay, as fixed-lag smoothing does, and differs in what it revises — it releases a confirmed track’s withheld prefix with coordinates, scores, and labels exactly as the detector produced them, so

the backward step changes set *membership*, not state.

Confidence in the track lifecycle. Recent 3D tracking has largely moved this decision from counting to confidence. CBMOT [1] replaces the minimum-hits criterion outright, propagating a tracklet score by constant decay, fusing it probabilistically with each matched detection score, and gating both output and termination on the refined score; it reports consistent AMOTA gains over count-based lifecycles. SimpleTrack [22] decomposes tracking-by-detection into pre-processing, association, motion model, and lifecycle management, and names *output* as a lifecycle decision separate from birth and death, letting low-score detections sustain a track through association without being emitted. Poly-MOT [19] and Fast-Poly [20] pair category-specific association with mixed count- and confidence-based termination and explicit control over which trajectories are written out; Fast-Poly adopts CBMOT’s score-fusion rule directly. Confidence-based temporal management is therefore established and well tuned, and we treat it as such: our contribution is not a new confirmation or lifecycle mechanism, and our associator is deliberately lightweight, so our absolute tracking scores are not comparable to these systems’. What we take from this line is a comparison rather than a competitor. Persistence-based confirmation and confidence accumulation are two temporal rules that both decide which detections reach the output, and both reduce how much is emitted; since detection-oriented metrics are strongly coupled to output volume, a comparison that does not hold that volume fixed reports the suppression and the rule together. Budget-matched comparison is routine when scoring detectors at a fixed proposal count; we apply it to temporal selection, holding fixed each of the two currencies such a rule spends — emitted detections and emitted identities — and read what separates the two rules under each.

Open-vocabulary 3D perception. OpenScene [23] distills 2D vision–language features into 3D points and LERF [13] embeds language into a radiance field; OpenMask3D [27] moves to instances via per-mask CLIP features over class-agnostic 3D proposals, and Open-YOLO 3D [5] replaces that extraction with 2D open-vocabulary boxes projected onto 3D masks. All four are *offline*, consuming a completed reconstruction. ConceptFusion [12] and Concept-Graphs [10] make the setting incremental, merging per-frame detections into persistent object nodes. A more recent line segments 3D instances fully online, computing proposals per frame: EmbodiedSAM [34] lifts per-frame 2D masks into 3D queries matched across views by a learned similarity, OnlineAnySeg [28] merges them into persistent instances zero-shot through voxel hashing, and [29] recasts online 3D segmentation as instance tracking with long-term query association. Cross-frame identity is thus an acknowledged design goal here, and in two of the three a trained objective, so we neither call it neglected nor rest any distinction

on being training-free — one of these systems is itself zero-shot. The gap is in what is measured: all are scored by per-scene instance-segmentation AP, the tracking-framed system included, so how often a persistent instance’s *class label* changes as observations accumulate is never quantified. Our indoor experiment measures that property, on a pipeline of this family, under a matched control.

Evaluation metrics. CLEAR MOTA [2] counts FP, FN and identity switches but is dominated by detection errors; IDF1 [24] rebalances toward identity by matching whole trajectories one-to-one. HOTA [21] was introduced because these conflate two capabilities, and factorises into a detection accuracy DetA and an association accuracy AssA whose geometric mean it is. That factorisation is what makes our result legible: detection and association quality can be read separately over an *identical* box set, precisely the distinction a temporal selection rule acts on. In 3D, nuScenes AMOTA [6] — following AB3DMOT’s integral formulation [32] — averages MOTA over recall thresholds and is class-aware, so it responds to labels the class-agnostic family ignores; we report both. Video consistency metrics (STQ [31], VPQ [14], TETA [17]) and open-vocabulary tracking benchmarks (TAO [9], OVTrack [18]) extend the family but require dense ground-truth tube identities over a fixed taxonomy. We use these metrics where computable rather than proposing a new one.

3. Method

Throughout this paper, *module* denotes a reusable post-processing operator composed of three existing parts — a class-agnostic associator, the standard confirmation test for track initialization, and an emission policy — applied to frozen detector output (Fig. 1). The module is deliberately minimal: no part of it changes a box’s coordinates, score, or label, and it decides only *which* detections enter the output stream and under *which* persistent identity. The confirmation test descends directly from tracking-by-detection practice [3, 32, 33]; the retrospective-emission completion and the controlled evidence for what the composition does (Sec. 4) are what we add.

3.1. Streaming pipeline

Per-frame detections (a 3D box with a score and a class label) are linked across frames into tracks by a class-agnostic nearest-centroid associator with a 2.0 m bird’s-eye-view center-distance threshold and a gap tolerance of 5 frames; no score threshold is applied before association. Association is performed either in the *sensor frame* (raw detector coordinates) or in the *world frame* (ego-motion-compensated global coordinates); both conditions are evaluated throughout, since ego-motion compensation changes how far the same physical object moves between frames and hence the difficulty of association. The associator is identical

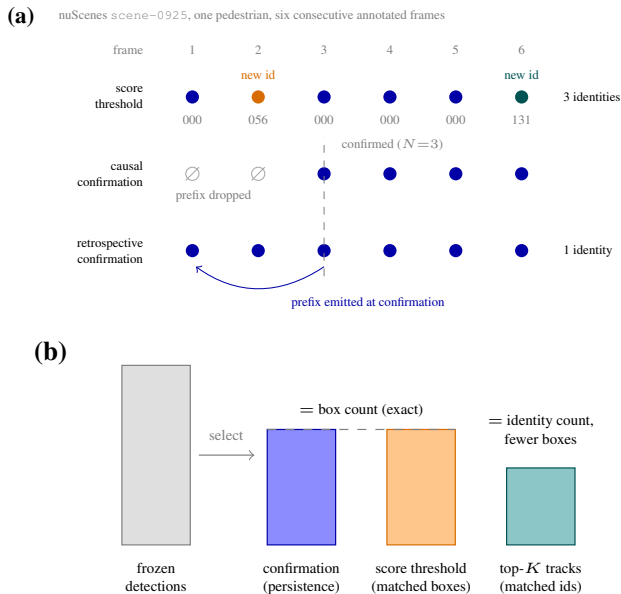


Figure 1. The module and the matched controls. (a) Emission policies on a measured example: one pedestrian tracked over six consecutive annotated frames of nuScenes scene-0925, with all three rows selecting from the same frozen detections. The emitted box is identical in all six frames, so the rows differ only in identity and in which frames are emitted. A per-frame score threshold carries no history forward and assigns three identities to the object (last three digits shown; two re-detections arrive as new identities). Confirmation-based initialization emits a track only once it has N observations ($N=3$): the *causal* policy drops the pre-confirmation prefix, so the object is emitted from frame 3 onward, while the *retrospective* policy emits the withheld prefix at confirmation time and recovers the complete trajectory under a single identity, at a latency of up to $N-1$ frames. (b) Every arm selects from the same frozen detection set. The detection-budget control is a score threshold calibrated to the module's exact box count; the identity-budget control keeps the module's track count per sequence, which necessarily emits fewer boxes because score-ranked tracks are shorter (Sec. 4.2).

tical in every arm of every experiment. It is deliberately lightweight rather than state of the art, because the quantity of interest is the difference a selection rule makes and not the highest attainable tracking score; absolute tracking numbers reported here are correspondingly lower than those of tuned nuScenes trackers and are not comparable to them.

3.2. Confirmation-based track initialization

A newly created track is *tentative*. It becomes *confirmed* once it has accumulated N observations; only confirmed tracks contribute detections to the output stream, and tracks that die before confirmation are never emitted. We use a single operating point, $N=3$, throughout the paper — the conventional setting in tracking-by-detection systems — rather

than tuning N per dataset or per metric. The module is therefore a pure temporal *selection*: the emitted detections are a subset of the baseline detector's output, chosen by track persistence instead of by score.

3.3. Emission policy

Confirmation leaves a policy choice for the first $N-1$ observations of a track that does get confirmed (Fig. 1a). A *causal* (online) policy emits nothing until confirmation and drops the pre-confirmation detections; an *offline retrospective* policy emits the withheld prefix at confirmation time, so a confirmed object's trajectory is complete from its first observation. The retrospective policy is only available to consumers that tolerate output latency: a detection may appear up to $N-1$ frames after it was observed — at the 2 Hz annotated-frame rate of our outdoor benchmark, 2 frames or 1.0 s for $N=3$. We adopt the retrospective policy for all main results and state this latency alongside every table; the causal policy is evaluated as an ablation (Sec. 5.3), since the two policies emit different numbers of boxes by construction.

4. Evaluation Protocol

The protocol is designed around one confound: confirmation reduces output volume, and the detection-oriented metrics this field reports are strongly coupled to how much a system emits, which makes a consistency gain hard to tell apart from the effect of suppression alone. Every claim in this paper is therefore made against a *matched control* — a non-temporal selection rule granted the same output budget — rather than against the unfiltered baseline.

4.1. Datasets and pipelines

Outdoor. The nuScenes [7] validation split, all 150 scenes at the 2 Hz annotated-frame rate. Detections come from a pretrained CenterPoint [36] detector whose per-frame output is computed once and *frozen*; every arm of every experiment replays the same stored detections (1,029,380 boxes over the split; mAP 0.3408, NDS 0.3150), so any metric difference between arms is attributable to the selection rule, not to detector variance.

Indoor. The ScanNet200 [25] validation split, all 312 scenes, in a streaming instance-segmentation pipeline: class-agnostic 3D instance-mask proposals from Mask3D [26] are labeled per frame by a 2D open-vocabulary detector [8], so each persistent instance receives a stream of per-frame class labels. The proposal set is likewise frozen across arms. This pipeline shares no detector, modality, or codebase beyond the temporal module with the outdoor one, which is the point: it tests whether the same confirmation rule transfers.

4.2. Matched controls

Two budgets are matched — emitted detections and emitted identities (Fig. 1b) — and under each we compare the module against a non-temporal score rule and against the other established temporal rule.

Detection-budget-matched control. A score threshold on the frozen baseline detections is calibrated so the control emits *exactly* as many boxes as the temporal module — 360,309 in the sensor frame (threshold 0.1875), 754,500 in the world frame (0.1211), both with zero count error. The match is exact because the count is a step function of the threshold whose breakpoints are the observed scores: the matched threshold is the K -th largest observed score, ties resolved upward so the count never exceeds K . This control asks whether temporal selection beats the detector’s own confidence ranking at equal detection volume.

Identity-budget-matched control. Per sequence, the control keeps the K highest-scoring tracks (ranked by mean detection score), K being the number of identities the temporal module emits there; counts match exactly (64,428 tracks sensor, 108,225 world). Five additional arms select the same K tracks *at random* (five seeds), scored on association accuracy, separating score-based selection from selection per se. The two budgets cannot hold at once: at equal track count the control emits fewer boxes than the module (190,708 vs. 360,309 sensor; 608,168 vs. 754,500 world), because score-ranked tracks are shorter than persistence-selected ones. We report this asymmetry and read the two controls jointly.

Confidence-accumulation control. To compare persistence against confidence rather than against a static score, a third control propagates a track score by constant decay and fuses it with each matched detection score, following CBMOT’s reported best update rule [1]; its parameters are transcribed from the authors’ released implementation, not tuned here, except that the gap tolerance is our associator’s rather than theirs, since the frozen association is shared by every arm. A single threshold on the refined score is calibrated to the same budget as the arm it is compared with, by the rule above, so this control is matched under both budgets. Its trained score-regression variant is not reproduced: it is fit on nuScenes and would give the control supervision no other arm receives.

Indoor control. The identity-budget design transfers directly: the control keeps the K highest-scoring instances per scene, where K matches the temporal module’s emitted-instance count (10,413 instances in total; the control realizes 10,410). Three additional arms select K instances *at random* per scene (three seeds), to separate the effect of score-based selection from that of selection per se.

Shared machinery. Baseline, temporal module, and every control share the frozen detections, the associator, and the evaluation code; they differ only in the selection rule.

4.3. Metrics

Outdoor. Official nuScenes mAP and NDS on the emitted box set; class-agnostic HOTA, AssA, DetA, detection recall (DetRe), and IDF1 via TrackEval [21, 24], with similarity $\text{sim} = \text{clip}(1 - d_{xy}/2.0, 0, 1)$ matching the associator’s 2.0 m threshold; and official class-aware AMOTA [6] over the 7 nuScenes tracking classes. The class-agnostic and class-aware families are reported side by side because they respond to different properties of the stream.

Indoor. The primary endpoint is the per-scene count of *label switches*: frames in which a persistent instance’s emitted class label differs from its label in the previous frame *in which it was emitted*. Counting against the last emission rather than the last frame is what makes the count comparable across arms, since the arms emit on different frames. Prediction stability has been measured separately from accuracy before — as fragment, position, and scale errors for video detection [37], and as a confidence/localisation stability index for 3D detection [30]: the former reports low correlation between its stability metric and the accuracy metric, the latter that stability “cannot be accessed by existing metrics such as mAP and MOTA” — but that family scores box trajectories against ground-truth tracks, which ScanNet200 does not provide, and none of it scores the *class* assigned to a persistent instance. We therefore define the switch count in that spirit for our setting. Because the arms also differ in how many observations they emit, we report the count both as a total and as a rate per switch opportunity, that is per emitted (instance, frame) observation after each instance’s first emission. The secondary endpoint is instance-segmentation AP on the final reconstruction, with the expectation of no material change.

4.4. Statistics

For all three main comparisons — detection-budget-matched, identity-budget-matched, and indoor — the arms, the primary endpoint, and the criterion for a negative result were fixed before the run in a pre-registration released with the code. For the outdoor comparisons that criterion was that the association gain hold in *both* association frames with scene-bootstrap intervals excluding zero, and that the dataset-level metrics (mAP, AMOTA) move consistently across the two association frames, whichever direction they move in; indoors, that the per-scene label-switch count fall. The causal-emission ablation (Sec. 5.3) predates that protocol and is what motivated it. All comparisons between the temporal module and a control are paired per scene. We report the per-scene Wilcoxon signed-rank test with the rank-biserial correlation as effect size, and a 95% bootstrap confidence interval on the difference of the aggregate metric obtained by resampling scenes (10,000 resamples, fixed seed 20260718). A difference is described as an improvement only when its interval excludes zero; inter-

vals that span zero are reported as no detectable difference, whatever their sign.

Intervals are available for the class-agnostic metrics, which are defined per scene and aggregate linearly over scenes. AMOTA is not, and neither are the nuScenes devkit metrics mAP and NDS: AMOTA integrates MOTA over recall thresholds estimated on the whole split, and mAP and NDS are likewise computed over the whole split, so a per-scene resample does not yield the same estimator. We report all three as point estimates without intervals. We therefore do not claim interval support for AMOTA anywhere; the evidence we offer for it is that its sign and rough magnitude are stable across four independent comparisons (two controls $\times$ two association frames).

5. Experiments

We report three outdoor comparisons on nuScenes val and one indoor comparison on ScanNet200 val, each between the confirmation-based module and a control granted the same output budget over shared frozen detections. Unless stated otherwise, outdoor results use retrospective emission, whose latency cost is $N-1$ frames (1.0 s at 2 Hz).

5.1. Detection-budget-matched comparison

Table 1 compares confirmation against controls calibrated to the identical box count, so the comparison is free of the volume confound: all matched arms emit 360,309 boxes in the sensor frame and 754,500 in the world frame, from the same frozen detector output.

Association accuracy improves. Against the threshold control AssA rises from 0.2044 to 0.2535 in the sensor frame and from 0.4175 to 0.4314 in the world frame, both intervals excluding zero ($[+0.0370, +0.0615]$ and $[+0.0086, +0.0195]$). Per scene the sensor-frame gain averages $+0.0340$ — a different estimator from the 0.0491 difference of aggregates — and holds in 110 of 150 scenes (Wilcoxon $p=4.6 \times 10^{-13}$, rank-biserial 0.68).

Class-aware tracking accuracy rises. AMOTA goes from 0.0547 to 0.0887 (sensor) and 0.1655 to 0.2033 (world), the largest relative effect we measure. AMOTA does not admit the per-scene resampling used for the other metrics (Sec. 4.4), so this is a point estimate; what we rely on is that it moves the same way in both frames and under the comparisons below.

Detection-oriented metrics fall. At the same budget, mAP drops $0.3324 \rightarrow 0.2023$ (sensor) and $0.3396 \rightarrow 0.2900$ (world), DetA drops $0.2184 \rightarrow 0.1537$ ($[-0.0700, -0.0593]$) and $0.1282 \rightarrow 0.1196$ ($[-0.0101, -0.0072]$), and detection recall drops $0.5182 \rightarrow 0.3859$ ($[-0.1483, -0.1170]$) and $0.5668 \rightarrow 0.5326$ ($[-0.0404, -0.0286]$); mAP is a dataset-level metric and is reported as a point estimate. Because HOTA is the geometric mean of DetA and AssA, the DetA loss dominates

and aggregate HOTA falls as well ($[-0.0180, -0.0103]$ sensor; $[-0.0059, -0.0027]$ world). The trade-off is explicit: the module buys association quality with detection quality rather than adding it on top.

IDF1 shows no detectable sensor-frame difference. The sensor-frame IDF1 interval spans zero ($[-0.0046, +0.0064]$; paired $p=0.45$, 77/150 scenes), so we report no difference there. In the world frame IDF1 is slightly *lower* for the confirmation module ($[-0.0091, -0.0047]$).

Against confidence accumulation, the two rules split the metrics. The accumulation control emits the same boxes and is the comparison that isolates *which* temporal rule is used rather than whether one is used at all. Confirmation attains higher AssA in both frames ($0.2217 \rightarrow 0.2535$ sensor, $0.4190 \rightarrow 0.4314$ world; intervals $[+0.0249, +0.0388]$ and $[+0.0089, +0.0163]$), and accumulation is ahead on every other measured axis, also with intervals excluding zero: HOTA ($[-0.0084, -0.0068]$ sensor, $[-0.0049, -0.0039]$ world), DetA ($[-0.0393, -0.0315]$, $[-0.0092, -0.0074]$), IDF1 ($[-0.0076, -0.0054]$, $[-0.0099, -0.0084]$) and DetRe. Neither rule dominates: they occupy different points of the association–detection trade-off, and the choice between them is a choice of which axis to spend. Because mAP was not computed for this control, we make no mAP-based claim about it.

Scope of the accumulation control. In the world frame this control is not statistically separable from the static threshold on any of the five metrics we resample (all intervals span zero); in the sensor frame it is. This concerns our reproduction under the frozen association and track-management configuration stated in Sec. 4.2, where the gap tolerance is our associator’s, and is not a claim about the original system or its published results.

5.2. Identity-budget-matched comparison

Matching the number of emitted identities rather than the number of boxes (Tab. 2) tests whether the association gain merely reflects having fewer identities to confuse. It does not. AssA remains higher for the confirmation module in both frames ($0.2346 \rightarrow 0.2535$ sensor, $0.4173 \rightarrow 0.4314$ world), with intervals excluding zero ($[+0.0129, +0.0258]$ and $[+0.0101, +0.0185]$) and per-scene wins in 117/150 and 111/150 scenes ($p=2.2 \times 10^{-16}$ and 4.1×10^{-11}). Selecting the same number of tracks *at random* rather than by score (Sec. 4.2) reproduces none of it: over the five seeded arms AssA reaches at most 0.2460 (sensor) and 0.3934 (world), both below the confirmation module (Tab. 2). AMOTA again favours confirmation ($0.0547 \rightarrow 0.0887$ sensor, $0.1587 \rightarrow 0.2033$ world).

The rule separation survives identity matching. Repeating the comparison against confidence accumulation with *identities* rather than boxes held fixed — the accumula-

Frame	Arm	Boxes	mAP	NDS	HOTA	AssA	DetA	DetRe	IDF1	AMOTA
Sensor	Baseline (unfiltered)	1,029,380	0.3408	0.3150	0.1372	0.1896	0.1007	0.5904	0.0759	0.0501
	Control (threshold)	360,309	0.3324	0.3110	0.2098	0.2044	0.2184	0.5182	0.1517	0.0547
	Control (accumulation)	360,309	—	—	0.2036	0.2217	0.1890	0.4601	0.1596	0.0701
	Confirmation ($N=3$)	360,309	0.2023	0.2518	0.1960	0.2535	0.1537	0.3859	0.1531	0.0887
World	Baseline (unfiltered)	1,029,380	0.3408	0.3150	0.2011	0.4089	0.0995	0.5838	0.1764	0.1580
	Control (threshold)	754,500	0.3396	0.3143	0.2307	0.4175	0.1282	0.5668	0.2190	0.1655
	Control (accumulation)	754,500	—	—	0.2307	0.4190	0.1278	0.5650	0.2212	0.1612
	Confirmation ($N=3$)	754,500	0.2900	0.3096	0.2263	0.4314	0.1196	0.5326	0.2121	0.2033

Table 1. **Detection-budget-matched comparison** on nuScenes val (150 scenes, frozen CenterPoint detections, retrospective emission). All three matched arms emit exactly the same number of boxes: a static score threshold (0.1875 sensor, 0.1211 world), a confidence-accumulation control in the manner of [1], and confirmation. At equal detection budget, confirmation attains the highest association accuracy (AssA) and class-aware tracking accuracy (AMOTA) in both frames, and the lowest detection-oriented accuracy (DetA, DetRe) and aggregate HOTA; accumulation lies between the two on the detection-oriented measures. Bold marks the best budget-matched arm and is omitted wherever the leading difference has a confidence interval spanning zero, which is sensor-frame IDF1 and, in the world frame, every column on which the threshold and accumulation controls are not separable. mAP and NDS were not computed for the accumulation control and are left blank rather than estimated. The unfiltered baseline is shown for reference only and is not budget-matched.

Arm	Boxes	HOTA	AssA	DetA	IDF1	AMOTA
<i>Sensor frame</i>						
Control (top- K)	190,708	0.2461	0.2346	0.2607	0.2081	0.0547
Control (rand.)	—	—	.2321–.2460	—	—	—
Confirmation	360,309	0.1960	0.2535	0.1537	0.1531	0.0887
<i>World frame</i>						
Control (top- K)	608,168	0.2538	0.4173	0.1553	0.2596	0.1587
Control (rand.)	—	—	.3878–.3934	—	—	—
Confirmation	754,500	0.2263	0.4314	0.1196	0.2121	0.2033

Table 2. **Identity-budget-matched comparison.** The control keeps, per scene, the K highest-scoring tracks, with K set to the number of identities the confirmation module emits (64,428 sensor, 108,225 world; exact match in both frames). The random- K rows (“rand.”) give the range over five seeds of the same selection made at random, scored on the claim-bearing metric (AssA) only. Matching identities leaves the box budgets unequal, in the control’s disfavour on volume-sensitive metrics; AssA and AMOTA nonetheless favour confirmation, in the same direction and of similar magnitude as in Tab. 1. The top- K control’s sensor-frame AMOTA coincides with the threshold control’s (Tab. 1) at this precision; the underlying values differ (0.054713 vs. 0.054715), verified against the raw outputs.

by mean instead of peak refined score, or matching identities dataset-wide instead of per scene, moves its AssA by less than 0.004, so the result is not an artifact of that choice. We report no AMOTA comparison under identity matching: the count of identities entering the class-aware evaluation is not equalised by matching the emitted count (Sec. 7).

Budget asymmetry. The two controls cannot be satisfied simultaneously. At equal identity count the control emits fewer boxes than the confirmation module (190,708 vs. 360,309 sensor; 608,168 vs. 754,500 world), because score-ranked tracks are shorter than persistence-selected ones. This works *against* the control on volume-sensitive metrics, so the AssA and AMOTA results here are conservative; conversely the control’s advantage on DetA and HOTA in this table is partly a budget artifact and should be read from Tab. 1, where boxes are matched exactly. IDF1 likewise favours the control here (Tab. 2), which the control’s shorter tracks help; we read it together with Tab. 1, where it shows no sensor-frame difference. The two tables are meant to be read together: the association gain survives both matchings, and the detection loss is established under the matching that isolates it.

5.3. Emission policy

The results above emit a confirmed track’s withheld prefix retrospectively, which costs 1.0 s of latency. Table 3 reports the causal alternative, in which the prefix is permanently dropped, again against a count-matched score threshold. The result splits by association frame. The AMOTA gain survives in both frames (0.0574 $\rightarrow$ 0.0800 sensor, 0.1825 $\rightarrow$ 0.2089 world), and in the sensor frame the association gain is not merely preserved but larger than under retrospective emission (0.2105 $\rightarrow$ 0.2861; interval [+0.0577, +0.0922] against [+0.0370, +0.0615] retrospectively, 110 of 150 scenes). In the world frame, however,

tion control ranked by its own refined score and cut at the confirmation module’s per-scene identity count, matched exactly — leaves the association separation unchanged: +0.0252 sensor ([+0.0200, +0.0316]) and +0.0172 world ([+0.0089, +0.0161]), against +0.0253 and +0.0170 at matched boxes. The two matchings agree to within 0.0003 in both frames, and the accompanying losses on HOTA, DetA, IDF1 and DetRe keep their sign and significance. Whichever currency is held fixed, then, the separation between the two rules is the same size, which excludes the simplest reading of it — that persistence gains association accuracy by carrying fewer identities. Ranking the control

Arm	mAP	HOTA	AssA	AMOTA
<i>Sensor frame</i>				
Control (threshold)	0.3213	0.2417	0.2105	0.0574
Confirmation, causal	0.1360	0.2056	0.2861	0.0800
<i>World frame</i>				
Control (threshold)	0.3367	0.2729	0.4315	0.1825
Confirmation, causal	0.2569	0.2456	0.4199	0.2089

Table 3. **Causal emission, detection-budget-matched** (221,297 boxes sensor at threshold 0.2519, 503,619 world at 0.1539; both arms exact). Dropping the pre-confirmation prefix instead of emitting it retrospectively preserves the AMOTA gain in both frames, but it reverses the world-frame association result: AssA is lower than the control, with a confidence interval entirely below zero. For this ablation we report confidence intervals for AssA only, the metric the comparison turns on; the remaining columns are point estimates.

causal emission *loses* association accuracy relative to the control (0.4315 $\rightarrow$ 0.4199), with a bootstrap interval entirely below zero ($[-0.0183, -0.0047]$; paired $p=0.008$, rank-biserial -0.25 , with confirmation ahead in only 53 of 150 scenes). The detection cost is larger throughout than under retrospective emission (mAP 0.3213 $\rightarrow$ 0.1360 sensor).

The latency-free policy is therefore not uniformly worse — it is worse where association was already easy. Ego-motion compensation removes most apparent motion between frames, so the world-frame control associates well on its own (AssA 0.4315 against 0.2105 in the sensor frame), and there the detections lost with the discarded prefix outweigh what confirmation cleans up. We offer that reading as interpretation; the operational consequence stands on the measurements alone: a system that cannot absorb $N-1$ frames of latency can still use confirmation, and should expect the association gain only if it associates in sensor coordinates.

5.4. Confirmation-window sensitivity

All results so far fix $N=3$. Figure 2 varies it over $\{2, 3, 4, 5\}$, giving each N its own exactly matched control, and asks what the choice is worth. $N=1$ is excluded: it confirms every track on its first observation, reproduces the unfiltered baseline detection-for-detection, and its matched control is therefore the same box set, so every difference is zero by construction. We use it only as an implementation check.

The association gain does not depend on the operating point. It is positive at every N in both association frames, and all eight intervals exclude zero. It also grows monotonically with N — in the sensor frame close to linearly (+0.0219, +0.0491, +0.0767, +0.1038), in the world frame with visible deceleration (+0.0058, +0.0139, +0.0184, +0.0220). The class-aware metric behaves differently: AMOTA flattens in the sensor frame (+0.0393 $\rightarrow$

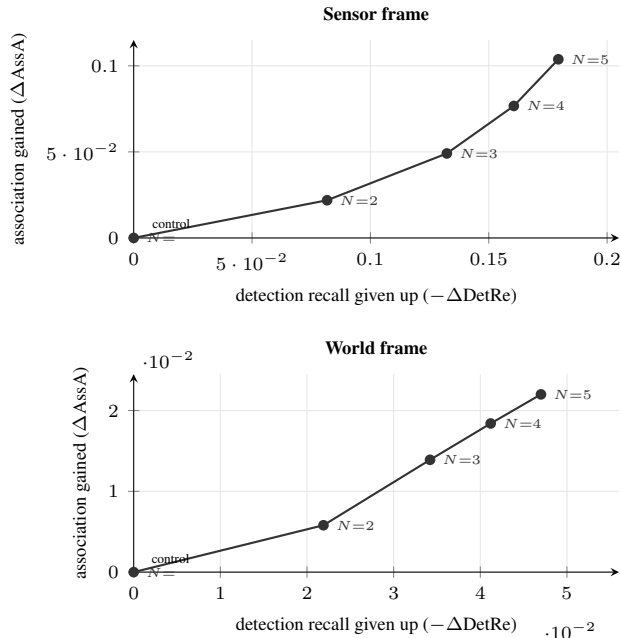


Figure 2. **Confirmation is a trade-off curve, not a tuned setting.** Each point is confirmation at that N minus its own detection-budget-matched threshold control, so the control sits at the origin. Moving right along the curve spends detection recall and $N-1$ frames of latency (0.5 s each at 2 Hz) and buys association accuracy; N is chosen by the latency a deployment can absorb rather than by a metric. The association gain is positive at every N in both frames and all eight bootstrap intervals exclude zero. Full numbers, including intervals and ΔAMOTA — which, unlike ΔAssA , turns over inside the range — are in the supplementary.

+0.0414 from $N=4$ to $N=5$) and in the world frame turns over, peaking at $N=4$ (+0.0424) and falling at $N=5$ (+0.0381). The detection cost, meanwhile, rises without turning over at any N we measured, and the latency rises linearly by construction.

No single N in this range is best on every metric, and $N=3$ is not the best on any of them. We report this rather than adjust the operating point: re-selecting N from the sweep would convert an inherited constant into a value tuned on the test metrics, the practice the matched controls exist to guard against. We keep $N=3$ because N is a latency budget, not a tuned hyperparameter: it is fixed by the $N-1$ frames of delay a deployment can absorb, and the role of the sweep is to price that budget — what each additional frame of delay buys in association and costs in detection. What the sweep establishes is that the consistency gain is a property of confirmation over the whole range tested, not an artifact of one setting, and that larger windows buy more of it at proportionally more detection cost and latency. The widening gap is driven by the confirmed arm rather than by a failing control, whose own AssA also rises as the matched

Arm	Inst.	Switches	AP	AP ₅₀
Baseline (unfiltered)	—	—	0.1956	—
Control (top- K)	10,410	23,303	0.1926	0.2504
Control (random K)	—	23,275–23,288	—	—
Confirmation ($N=3$)	10,413	17,045	0.1954	0.2532

Table 4. **Indoor comparison** on ScanNet200 val (312 scenes, frozen Mask3D proposals with per-frame open-vocabulary labels). The primary endpoint is the per-scene count of label switches (lower is better); the control keeps the K highest-scoring instances per scene with K matched to the confirmation module’s emitted-instance count. The random- K row spans three seeds (23,275, 23,277, 23,288) and is scored on the primary endpoint only. Bold marks the primary endpoint; the AP columns are secondary and are not tested. The unfiltered baseline is the frozen validity anchor, for which only AP was recorded; it is not an arm of the comparison, hence its empty cells.

budget tightens (0.1980 $\rightarrow$ 0.2107 sensor, 0.4131 $\rightarrow$ 0.4262 world).

5.5. Indoor: semantic stability

The indoor experiment asks whether the same rule improves *semantic* stability in a pipeline that shares no detector, modality, or codebase with the outdoor one. Table 4 reports the primary endpoint. Against the identity-matched control, confirmation reduces label switches from 23,303 to 17,045, a 26.9% reduction at an essentially equal instance budget (10,413 vs. 10,410). Per scene the reduction is 18.0 switches at the median (20.1 on average, 95% CI on the mean [18.7, 21.5]) and holds in 309 of 312 scenes (Wilcoxon $p=1.5 \times 10^{-52}$, rank-biserial 0.99).

The three random- K arms (23,275, 23,277, 23,288 switches) are indistinguishable from the score-ranked control and far from the confirmation module, so the reduction is specific to temporal selection rather than to emitting fewer instances by any rule.

The secondary endpoint behaves as expected: instance segmentation AP is 0.1954 against 0.1956 for the unfiltered baseline and 0.1926 for the control, i.e. no material change. Unlike the outdoor setting, the indoor gain therefore does not come at a measurable detection-quality cost — the two pipelines differ in whether the withheld detections carry segmentation mass, and we report the difference in effect without claiming the outdoor trade-off is avoidable in general.

6. Discussion

What the module improves, and what the controls add. Compared against an unfiltered baseline, the module appears to improve nearly every tracking metric. Compared against a non-temporal rule with the same output budget, part of that apparent improvement turns out to be the ef-

fect of emitting less, and what remains is a sharper claim: association accuracy and class-aware tracking accuracy improve, detection-oriented accuracy degrades, and identity F1 is unchanged at a matched detection budget but lower at a matched identity budget, where the control’s shorter tracks favour it. Against the other temporal rule the picture is not an ordering but a split: confidence accumulation keeps the detection- and identity-oriented measures, persistence takes association accuracy, and both halves of that statement have intervals excluding zero under either budget. The controls are what let us say any of this; they are the evidence standard behind the claim rather than a separate finding, and the general lesson — a component that spends output volume must be compared at fixed volume, including against other components that spend it — is available to anyone evaluating a similar module.

Association and detection respond to opposite things.

What a confirmation rule removes is fixed by construction: tracks observed fewer than N times. The falling detection recall shows that some of those withheld detections matched ground truth, which is the cost side. The gain side admits an interpretation we can offer but did not measure separately — that the identities surviving confirmation are the ones an associator handles most reliably. What the measurements establish is that the two halves move in opposite directions over an identical box set, which is why we report DetA and AssA rather than HOTA alone: the aggregate hides a trade-off it averages over.

The two domains agree in direction, not in cost. Outdoors the identity gain is paid for in detection quality; indoors the analogous semantic-stability gain leaves AP unchanged. In both pipelines temporal selection improves a property the per-frame metric does not measure — identity consistency outdoors, label stability indoors — across two systems sharing nothing but the rule. The differing cost is a property of the pipelines, and we do not generalise from the indoor result that the outdoor cost is avoidable.

Practical reading. Within the settings we measured, the case our evidence favours is a system whose output is consumed as persistent objects: a map, a scene graph, a planner’s object memory. A system scored on per-frame detection quality should not adopt confirmation-based initialization, and one that needs the detection- and identity-oriented measures is better served by accumulating confidence than by counting observations. Latency decides not whether to adopt confirmation but what to expect: with $N-1$ frames of slack the association gain holds in both frames, and without it only in the sensor frame (Sec. 5.3).

7. Limitations

One tracking-GT dataset. The full tracking evaluation (HOTA family, AMOTA) is nuScenes only. ScanNet200 has no instance-tracking ground truth, so the indoor endpoint is

label switching rather than an identity metric, and the two domains are not measured with the same instrument. Multi-dataset tracking validation (Waymo, Argoverse) remains future work.

One operating point. We fix $N=3$ and do not tune it per dataset or metric. The value is inherited: it is the conventional default in tracking-by-detection systems [3, 33], not a setting we chose by inspecting our own results, and the same value is used in both domains. This avoids selection on the test metric. The outdoor sweep of Sec. 5.4 shows the association gain holds across $N \in \{2, 3, 4, 5\}$, so the claim is not specific to $N=3$; it also shows $N=3$ is not the best setting on any single metric, and that the class-aware metric turns over inside the range while the class-agnostic one does not. N is therefore chosen as a latency budget rather than tuned. We did not sweep N indoors, and we do not know where either curve ends beyond $N=5$.

Budget matching is not simultaneous. Matching detections leaves identity counts unequal and vice versa (Sec. 5.2). We report both matchings and their directions of bias rather than a single comparison that is matched in both currencies, which does not exist for a rule that changes track lengths. Matching emitted identities also does not equalise the identities that reach the class-aware evaluation, which scores a subset of classes: at an identical emitted count the accumulation control carries 13% more identities into that evaluation in the world frame. We therefore restrict the identity-matched comparison to the class-agnostic metrics and report AMOTA only under detection matching.

One reproduced comparator. The confidence-accumulation control is our reimplement, run inside our frozen association with our gap tolerance and without the comparator’s trained score variant. It is a faithful control for the mechanism, not a reproduction of a published system, and no statement here should be read as characterising that system’s own performance. mAP was not computed for this control, so the detection-side comparison against it rests on DetA and DetRe.

Latency. All main results use retrospective emission and are therefore not achievable online at zero latency. The causal variant is measured (Sec. 5.3) and preserves the association gain only in the sensor frame.

Frozen detector. Every arm replays one stored detection set, which is what makes the comparison clean, but it also means we characterise the rule’s effect for a single detector’s error profile. A detector with fewer short-lived false positives has less for confirmation to remove.

8. Future Work

Because the causal variant forfeits the world-frame association gain and the retrospective one buys it with $N-1$ frames of latency, the interesting design space is between them: a policy that emits provisionally and revises, or

that varies confirmation depth with association confidence, would trade latency against identity consistency continuously rather than at the two endpoints we measure. A rule combining persistence with accumulated confidence is the natural companion question, given that the two split the metrics here. The controls of Sec. 4 support these measurements unmodified, as they do a stronger associator or a different detector.

9. Conclusion

We presented a training-free retrospective confirmation module for streaming 3D perception: detections are linked into tracks, a track enters the output stream only after N observations, and its withheld prefix is then emitted retrospectively. It requires no retraining, leaves box coordinates, scores, and labels untouched, and runs at one operating point across two pipelines that share nothing else. On nuScenes with a frozen detector it improves association accuracy and class-aware tracking accuracy in both association frames; on ScanNet200 it reduces class-label switching by 26.9% with segmentation AP unchanged. These results are established against controls granted the module’s own output budget — in detections and in identities — so what they credit is temporal selection rather than the smaller volume it emits. The gain is bought rather than added: at an equal detection budget the module trades away per-frame detection quality (Sec. 5.1), and the retrospective step costs $N-1$ frames of latency (Sec. 5.3). Held against the other established temporal rule at the same budgets, confirmation does not dominate it: accumulating confidence retains the detection- and identity-oriented measures while counting observations takes association accuracy, and that separation is the same size whether detections or identities are the quantity held fixed. What a system should spend its output budget on therefore depends on which axis it is consumed along — and any component that changes how much a system emits can be held to the same standard with the same controls.

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